

MATHEMATICS — CLASS X

LINEAR EQUATIONS IN TWO VARIABLES

Unit Test | Session 2026–27 | CBSE Syllabus

SECTION A — Multiple Choice Questions ($1 \times 10 = 10$ Marks)

Choose the correct answer. Each question carries 1 mark.

- Q1.** The pair of linear equations $2x + 3y = 5$ and $4x + 6y = 10$ has: **[[1]]**
(a) No solution (b) Exactly one solution (c) Infinitely many solutions (d) Exactly two solutions
- Q2.** For what value of k will $kx + 3y = k - 3$ and $12x + ky = k$ have infinitely many solutions? **[[1]]**
(a) $k = 6$ (b) $k = -6$ (c) $k = 3$ (d) $k = -3$
- Q3.** The graphical representation of a pair of equations that has NO solution is: **[[1]]**
(a) Coincident lines (b) Intersecting lines (c) Parallel lines (d) Perpendicular lines
- Q4.** If the system $x + 2y = 3$ and $5x + ky = 15$ has a unique solution, then $k \neq$: **[[1]]**
(a) 5 (b) 10 (c) 6 (d) 15
- Q5.** The solution of the equations $x + y = 14$ and $x - y = 4$ is: **[[1]]**
(a) $x = 9, y = 5$ (b) $x = 5, y = 9$ (c) $x = 7, y = 7$ (d) $x = 8, y = 6$
- Q6.** The condition $a_1/a_2 = b_1/b_2 \neq c_1/c_2$ for a pair of linear equations represents: **[[1]]**
(a) Unique solution (b) No solution (c) Infinitely many solutions (d) Cannot be determined
- Q7.** The point of intersection of the lines $x = 4$ and $y = 3$ is: **[[1]]**
(a) (3, 4) (b) (4, 3) (c) (0, 3) (d) (4, 0)
- Q8.** If $2x + y = 5$ and $x + y = 3$, then $x =$: **[[1]]**
(a) 1 (b) 2 (c) 3 (d) 4
- Q9.** Which of the following is NOT a linear equation in two variables? **[[1]]**
(a) $2x + 3y = 7$ (b) $x^2 + y = 4$ (c) $x - y = 0$ (d) $3x = 5 - 2y$
- Q10.** The pair of equations $y = 0$ and $y = -5$ has: **[[1]]**
(a) One solution (b) Two solutions (c) Infinitely many solutions (d) No solution

SECTION B — Very Short Answer Questions ($2 \times 6 = 12$ Marks)

Each question carries 2 marks. Show necessary working.

- Q11.** Check whether $3x + 2y = 5$ and $2x - 3y = 7$ is consistent or inconsistent. **[[2]]**
- Q12.** For what value of p will $px + 2y = 7$ and $3x + y = 5$ have a unique solution? **[[2]]**
- Q13.** Find the values of a and b for which $2x + 3y = 7$ and $(a + b)x + (2a - b)y = 21$ has infinitely many solutions. **[[2]]**
- Q14.** Find the value of k so that the point $(k, 2 - k)$ lies on the line $x + y = 6$. **[[2]]**
- Q15.** The sum of digits of a two-digit number is 9. On reversing the digits the number decreases by 27. Form a pair of linear equations. (Do not solve.) **[[2]]**
- Q16.** Solve for x and y : $99x + 101y = 499$ and $101x + 99y = 501$. **[[2]]**

SECTION C — Short Answer Questions – I ($3 \times 5 = 15$ Marks)

Each question carries 3 marks. Internal choices are indicated.

Q17. Solve by the ELIMINATION method: $3x - 5y - 4 = 0$ and $9x = 2y + 7$. **[[3]] OR** Solve by SUBSTITUTION: $x/2 + 2y/3 = -1$ and $x - y/3 = 3$

Q18. Five years hence, the age of Jacob will be three times that of his son. Five years ago, Jacob's age was seven times that of his son. Find their present ages. **[[3]]**

Q19. Places A and B are 100 km apart. One car starts from A and another from B at the same time. If they travel in the same direction they meet in 5 hours; if they travel towards each other they meet in 1 hour. Find the speeds of the two cars. **[[3]]**

Q20. For what values of p and q will $4x + 5y = 2$ and $(2p + 7q)x + (p + 8q)y = 2q - p + 1$ have infinitely many solutions? **[[3]] OR** Find the value of k for which $3x - ky = 7$ and $6x - 10y = 3$ has no solution.

Q21. Draw the graph of $x - y + 2 = 0$ and $4x - y - 4 = 0$. Find the vertices of the triangle formed with the x-axis and calculate its area. **[[3]]**

SECTION D — Short Answer Questions – II ($4 \times 4 = 16$ Marks)

Each question carries 4 marks. Internal choices are provided.

Q22. Solve the following pairs of equations (use $u = 1/x$ and $v = 1/y$ wherever applicable): **[[4]]**

(a) $2/x + 3/y = 13$ and $5/x - 4/y = -2$

(b) $1/(3x+y) + 1/(3x-y) = 3/4$ and $1/[2(3x+y)] - 1/[2(3x-y)] = -1/8$

OR Solve graphically: $2x + y = 6$ and $2x - y + 2 = 0$. Find the ratio of areas of the two triangles formed with the x-axis.

Q23. A boat goes 30 km upstream and 44 km downstream in 10 hours. In 13 hours it goes 40 km upstream and 55 km downstream. Find the speed of the stream and the speed of the boat in still water. **[[4]]**

Q24. The ratio of incomes of two persons is 9 : 7 and the ratio of their expenditures is 4 : 3. If each saves Rs. 2000 per month, find their monthly incomes. **[[4]] OR** A man travels 600 km partly by train and partly by car. It takes 8 h 40 min if he travels 320 km by train and the rest by car. It would take 30 min more if he travels 200 km by train and the rest by car. Find the speed of the train and car.

Q25. Students of a class are made to stand in rows. If 3 students are extra in a row, there is 1 row less. If 3 students are fewer in a row, there are 2 rows more. Find the number of students in the class. **[[4]]**

SECTION E — Long Answer Questions ($5 \times 3 = 15$ Marks)

Each question carries 5 marks. Show all steps clearly. Internal choices are provided.

Q26. Solve the following pairs of equations by reducing them to linear form: **[[5]]**

(a) $1/(2x) + 1/(3y) = 2$ and $1/(3x) + 1/(2y) = 13/6$ (b) $6x + 3y = 6xy$ and $2x + 4y = 5xy$

OR A train covered a certain distance at uniform speed. Had the train been 10 km/h faster, it would have taken 2 hours less. Had it been 10 km/h slower, it would have taken 3 hours more. Find the distance covered.

Q27. A lending library charges a fixed fee for the first 3 days and an additional charge per day thereafter. Saritha paid Rs. 27 for a book kept 7 days; Susy paid Rs. 21 for a book kept 5 days. Find the fixed charge and the daily extra charge. **[[5]]**

OR The area of a rectangle is reduced by 9 sq. units if the length is decreased by 5 and breadth increased by 3. It increases by 67 sq. units if length increases by 3 and breadth increases by 2. Find the dimensions.

Q28. Read the case study carefully and answer the parts that follow: **[[5]]**

Amit bought 3 notebooks and 2 pens for Rs. 80. His friend Sumit bought 4 notebooks and 3 pens for Rs. 110 from the same shop. The shopkeeper said that the cost of one pen is the same as the cost of one notebook.

- (i) Form a pair of linear equations from the above situation.
- (ii) Find the cost of one notebook and one pen by any algebraic method.
- (iii) If Riya wants to buy 5 notebooks and 4 pens, how much will she need to pay? Also, what amount will she save if she gets a 10% discount?

OR The cost of 5 oranges and 3 apples is Rs. 35. The cost of 2 oranges and 4 apples is Rs. 28. Find the cost of each. Then find the total cost of 1 dozen oranges and 2 dozen apples.
